SkeletKey

First Folding, Electric, and Modular Motorcycle

Master Operations & Documentation Index

10,000 Units / Year · 40 Units / Day · 100% Consigned Assembly · 6061-T6 Chassis

SkeletKey is a high-performance folding electric motorcycle engineered for real urban mobility: a rigid 6061-T6 aluminum chassis, modular quick-swap battery pack, and sustained highway-capable speeds with a verified range exceeding 100 miles. The market currently forces a painful trade-off—true 100-mile electric motorcycles start at roughly $10,000, and the closest competitors that approach this range are still $3,000 more expensive than SkeletKey's $7,000 target. Those alternatives do not fold, lack a modular battery architecture, and cannot be easily transported or stored in apartments, elevators, or public transit. SkeletKey closes the gap by delivering the same long-range capability with only a modest 10 mph reduction in top speed, while adding true foldability, modular power, and travel-friendly dimensions that no current competitor offers at this price point.

This master index consolidates every operational framework, technical specification, financial model, quality protocol, supplier template, and logistics document for the SkeletKey program. Each entry lists the source filename and a one-sentence purpose so teams can instantly locate the authoritative reference for engineering, commercial, quality, or capital decisions. Click any document name to open the matching file in the Documents folder. The same files are also embedded as PDF attachments (paperclip panel) for fully offline use.

1 · PROTOTYPE SCHEMATICS & DESIGN PLAN

Puca Schematics and Design Plan.md.txt	Comprehensive technical build and integration blueprint for the SkeletKey high-performance folding prototype, covering isolated exoskeleton power cradle, 60.8V 16S2P electrical schematic, aerodynamics for 100-mile range, controller thermal programming, cost-optimized $2,839 BOM, and dropout geometry enabling stable 70 mph operation.

2 · STRATEGIC & BUSINESS PLANNING

Master Business Plan Summary.md	Executive summary of the asset-light consigned manufacturing model, three-layer QA firewall, unit economics delivering 47% gross margins, and non-dilutive financing structure for the 10,000-unit program.
Investor Pitch Deck Outline.md	Complete eight-slide investor pitch outline covering mission, CapEx avoidance, 6061-T6 engineering, consigned manufacturing engine, QA firewall, disruptive unit economics, and capital ask.

3 · TECHNICAL ENGINEERING & SPECIFICATIONS

Technical Fact Sheet.md	Official technical appendix detailing 6061-T6 chassis metrics (advantages over base 6061 temper), GTAW/GMAW welding protocols, post-weld heat treatment, 16S2P powertrain architecture, and the mandatory three-layer manufacturing quality criteria for high-speed stability.
Engineering Change Notice.md	Formal ECN-2026-001 standardizing all primary structural frame components to certified heat-treated 6061-T6 aluminum for superior yield strength and resistance to high-speed flex/wobble, including affected systems, QA validation rules, and inventory disposition strategy.

4 · MANUFACTURING ARCHITECTURE & QUALITY ASSURANCE

Assembly Strategy.pdf	Defines the consigned assembly model, tiered production decoupling (Tier-2 structural/powertrain cells feeding a Tier-1 final marriage hub), and JIT logistics required to sustain 40 completed vehicles per working day.
QA Strategy.pdf	Establishes the independent three-layer QA firewall: Layer 1 ultrasonic/X-ray on 6061-T6 welds, Layer 2 dielectric isolation and thermal cycling of 16S2P packs, Layer 3 mandatory 10-minute dynamometer plus high-pressure water-spray validation.
Due Diligence Checklist.md	Facility site-visit protocol checklist verifying post-weld aging ovens, NDT bays, ESD-protected battery lines, inline insulation testers, thermal chambers, dynamometer stations, and environmental spray booths.
Kickoff Meeting Agenda.md	Timed factory kickoff agenda covering production cadence targets (40 units/day), 6061-T6 engineering standards, three-layer QA integration, and tooling/die delivery action items.
Action Item Tracker.md	Post-meeting tracker assigning owners and deliverables for CAD release, 30% tooling down-payment, UT calibration, dynamometer installation, EDI/MRP linkage, and asset-backed inventory financing activation.
Production Milestone Gantt.md	Week-by-week Gantt framework spanning tooling & die cutting, pre-production verification, regulatory certification (DOT/UN38.3), and full-scale ramp to steady-state 40 units/day.

5 · FINANCIAL MODELS, TOOLING & CASH FLOW

Financial Reality Check.pdf	Unit economics matrix ($7,000 MSRP / $3,200 BOM / $500 assembly / $3,300 margin) and allocation of the $33 million annual gross-profit pool to logistics, regulatory compliance, and non-dilutive inventory financing.
BOM Cost Matrix.md	High-level Bill of Materials cost-tracking matrix by component category targeting $3,200 unit cost at 10,000-unit volume, with material-price variance thresholds and Tier-2 yield-rate controls.
Cash Flow Pro Forma.md	Quarterly cash-flow staging from Q1 tooling/LRIP through Q3–Q4 peak velocity, emphasizing inventory financing lines structured solely against product collateral and omitting personal guarantees.
Tooling Amortization Schedule.md	Amortization schedule absorbing $300,000 in custom 6061-T6 extrusion dies ($120k) and stamping tooling ($180k) into temporary per-unit surcharges ($24 + $18) with full ownership transfer at volume milestones.

6 · SUPPLIER ENGAGEMENT, NEGOTIATION & CONTRACTING

RFP Template.md	Complete Request-for-Proposal template for Tier-1 mobility contract manufacturers defining the consigned model, 40-unit/day throughput, tooling ownership retention, and strict three-layer QA parameters.
Vendor Outreach Email.md	Initial outreach email template introducing the 10,000-unit consigned program, $500/unit assembly allocation, 6061-T6 and 16S2P technical thresholds, and capacity/quality infrastructure inquiry questions.
Letter of Intent.md	Non-binding LOI template establishing production run-rate, 100% consignment, $500/unit labor target, three-layer QA firewall, scrap caps, and contractual omission of founder personal guarantees.
Supplier Negotiation Playbook.md	Ready counter-strategies for material-handling fee demands, tooling CapEx pushback, and multi-month capacity retainers while protecting the $500 assembly ceiling and working-capital liquidity.

7 · LOGISTICS, FREIGHT & WAREHOUSE OPERATIONS

Freight Forwarder RFP.md	RFP for global freight forwarders managing inbound 6061-T6 chassis extrusions and Class-9 16S2P powertrain packs, synchronized to a 200-component-set / 5-day injection cadence with required telematics and free-time SLAs.
Warehouse Operations Guide.md	Finished-vehicle warehouse playbook covering inbound unboxing inspection of 6061-T6 frames and 16S2P packs, high-density staging, Class-9 HazMat enclosure, 50% SoC outbound charging, and real-time inventory financing telemetry.
Carrier Scorecard and Safety Outline.pdf	Weighted carrier performance scorecard (OTD 30%, transit 20%, cargo care 20%, tracking 15%, billing 15%) plus a four-module warehouse safety training outline (facility hazards, ergonomics, PIT/forklift, emergency response).
Master Logistics Operations Index.pdf	Earlier partial logistics index (now superseded by this comprehensive master document) that originally referenced the carrier scorecard and warehouse safety outline.

Document Control · All source files reside in the program repository. This index is the single entry point for the authoritative version of each framework. Update whenever a new operational template, ECN, or financial model is released. Primary structural standard: heat-treated **6061-T6 aluminum**. Primary powertrain: **16S2P NMC**. Commercial model: **100% Consigned Assembly** with complete omission of founder personal guarantees.